Nikolaos Louloudakis, Ph.D.

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Professional Experience

Barclays

AI Researcher & Engineer (AVP) | October 2025 - Present

Conducting AI-oriented R&D as a member of the distinguished engineering team on AI.

· Applying automated analysis, evaluation, and testing of AI agents and LLM models..

· Enabling research collaborations with academia and enhancing AI-oriented information retrieval.

· Developing and evaluating observability solutions, sharing expertise, and guiding stakeholders.

University of Edinburgh

Research Collaborator, AI Testing | October 2025 - Present

Postdoctoral Researcher, AI Testing | October 2024 - October 2025 | 2 Top-Tier Publications (ICSE, CC)

Worked on differential and metamorphic testing of AI/ML models and compilers.

· Applied research on DNN optimizer fault detection, discovering 14 unknown bugs in ONNX Optimizer.

· Implemented a utility that allows selective model quantization and deployment.

· Co-supervised an MSc student and a research intern that worked on AI systems research.

· PC: AAAI (2024), ICSE DeepTest, AIware, TOSEM, NEUNET, EMSE (2026), ICSE AGENTS (2027)

Oracle Labs

Research Associate (Intern) | May 2024 – August 2024

Implemented a novel feature for Sandwood, an ML compiler for probabilistic models.

Carbon Glance

Co-Founder & CTO | January 2022 – May 2024

Coordinated the design and development of the company product, related to carbon regulations.

· Formed and led a team of 5 talented and strong software engineers.

· Contributed in key decisions, and introduced procedures towards effective product development.

Workable

Software Engineer | September 2018 – April 2021 | 🏆 *Workable Security Champion*

Designed, developed, and integrated services and components for an Applicant Tracking System, following the full software development lifecycle while collaborating with cross-functional teams.

· *AMP:* Implemented a compiler to automatically create hybrid, valid AMP pages from React.

· *CI/CD:* Implemented the testing pipeline for the brand design assets of the company.

· *Service Integration:* Developed custom third-party search and i18n resource integrations.

· *Analytics:* Refined the analytics integration mechanism for all company services.

PeopleCert Education

Coding Bootcamp Instructor | February 2018 - June 2021 (Various Iterations)

Served as coding instructor to professionals of ages 20 - 50, on multiple bootcamp iterations.

Accenture

Software Engineer - Consultant | March 2018 – August 2018

Worked in the largest telecommunication vendor in Greece.

· Integrated services in pipelines affecting procedures related to thousands of customers.

· Developed automation utilities that increased team productivity, saving valuable team man-hours.

Hellenic Army

Software Engineer - Analyst | June 2017 – February 2018

Worked on 2 .NET-based Content Management Systems.

ICS-FORTH Research Foundation

Software Engineer - Graduate Research Associate | *July 2014 – April 2017*

🏆 *Graduate Scholarship* | *Best Paper Award* | *3 Publications*

Applied Research & Development on IoT environments and automation of testing.

· Implemented the database infrastructure and a number of modules for a smart educational suite.

· Contributed towards the implementation and testing of a programmable IoT environment.

StackOverflow (Volunteer)

Established Member | *September 2013 – October 2023*

SO Profile | 220+ answers | Estimated impact: 4M Users | 6K reputation points | Top 7%

Education

Ph.D., Computer Science (AI Testing), University of Edinburgh

January 2021 – November 2024 | *First-authored 5 Top-Tier Venue Publications*

Thesis: *Testing Deep Neural Networks Across Different Computational Configurations*

Supplementary Contributions: *TA in Software Testing (4 semesters, 150+ students each) | Reviewer at AAAI SRRAI track | Supervisor of 4 M.Sc. theses | Leader of a M.Sc. Beyond-Curriculum Initiative*

M.Sc., Information Systems & Human-Computer Interaction, University of Crete

March 2014 – April 2017, 9.13/10 (Excellent) | *1 Publication (+ 3 more with ICS-FORTH)*

Thesis: *AmITest: A Framework for Semi-Automated Testing of Ambient Intelligence Environments*

TA: Human – Computer Interaction (HCI), Advanced Topics in HCI

B.Sc., Computer Science & Software Engineering, University of Crete

October 2008 – February 2014, 7.49/10 (Top 3% Of Class 2008)

Thesis: *VHF Transceiver UI Components of a System for Emergency Situations*

Skills

Python, JavaScript, C/C++, Java, C#, Ruby, SQL, ONNX, Apache TVM, PyTorch, Keras, PostgreSQL, MongoDB, Express.js, React, Node.js, Entity Framework, Playwright, Jest, Mocha, JUnit, Git, LLMs, Agents, CNNs, Analytical Thinking, Research, Testing, Validation, AI Systems, UX Design, Code Generation, Compilers

Selected Publications

A Selective Quantization Tuner for ONNX Models

N. Louloudakis, A. Rajan | *In IEEE/ACM ICSE NIER 2026, Rio de Janeiro, Brazil*

DiTOX: Fault Detection and Localization in the ONNX Optimizer

N. Louloudakis, A. Rajan | *In ACM SIGPLAN CC 2026, Sydney, Australia (& talk at ONNX Meetup 2026)*

FetaFix: Automatic Fault Localization and Repair of Deep Learning Model Conversions

N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In EASE 2025, Istanbul, Turkey*

Exploring Robustness of Image Recognition Models on Hardware Accelerators

N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In IEEE ICST Mutation 2025, Naples, Italy*

DeltaNN: Differential Testing to Evaluate Robustness of Image Recognition Models

N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In IEEE ICSME Industry Track 2023, Bogotá, Colombia*

Fault Localization for Buggy Deep Learning Framework Conversions in Image Recognition

N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In IEEE/ACM ASE NIER Track 2023, Kirchberg, Luxembourg*

Assessing Robustness of Image Recognition Models to Changes in the Computational Environment

N. Louloudakis, P. Gibson, J. Cano, A. Rajan | *In NeurIPS ML Safety 2022, New Orleans, USA*